CHUKA UNIVERSITY
Center for Data Analytics & Modeling
Website: https://cdam.chuka.ac.ke

Session ten: Capstone Projects in python

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Capstone Project 1: Retail Sales Performance Analysis

ProjectLink: https://drive.google.com/file/d/1Qiph-9zCeKviL5pxjWZWddn5cJaOM42t/view?usp=drive_link

DatasetLink:
https://drive.google.com/file/d/1q2HHuR3ZTZaCXDLDCoEjwM85bHJ3XEBH/view?usp=drive_link

Project Scenario

Background

ABC Retail Stores Ltd. is a medium-sized retail company with branches in the North, South, East, and West regions. The company sells a variety of consumer products, including electronics, furniture, office supplies, and clothing. Over the past year, thousands of sales transactions have been recorded through the company's point-of-sale system.

Although the organization has accumulated substantial operational data, management has not conducted a comprehensive analysis to evaluate sales performance, customer purchasing behavior, regional trends, or product category performance. Furthermore, preliminary reviews indicate that the dataset contains missing values, duplicate records, inconsistent category names, and data entry errors that must be resolved before meaningful analysis can be performed.

You have recently joined the company as a Junior Data Analyst. Your responsibility is to clean, analyze, and visualize the retail sales data using Python and prepare an analytical report that supports managerial decision-making.

Business Problem

Senior management requires answers to several operational questions, including:

- Which regions generate the highest sales revenue?

- Which product categories contribute the most to total sales?
- What are the monthly sales trends?
- How do customer demographics vary across regions?
- How frequently are discounts offered?
- Are there inconsistencies within the sales records?
- Which products are purchased most frequently?

Without reliable descriptive analysis, managers cannot effectively monitor business performance, evaluate marketing strategies, or optimize inventory planning.

Importance of the Analysis

The analysis will enable management to:

- Improve data quality by identifying and correcting errors.
- Understand regional sales performance.
- Evaluate product category performance.
- Analyze customer demographics.
- Examine discount usage.
- Produce informative statistical summaries and visualizations.
- Support evidence-based operational decisions.

Expected Outcome

At the end of the project, management expects a professional analytical report summarizing:

- Regional sales performance
- Product category performance
- Customer demographics
- Monthly sales trends
- Discount analysis
- Descriptive statistical summaries
- Professional visualizations
- Practical business recommendations

Capstone Project 2: Customer Banking Transactions Analysis

ProjectLink: https://drive.google.com/file/d/1X-5XudKpAkbUQQUpo2GCGXPvUIYUXk0A/view?usp=drive_link

DatasetLink: https://drive.google.com/file/d/15gHvGQv2x2gPP4ydaYUrqvaM33fTYfsl/view?usp=drive_link

Project Scenario

Background

SecureBank Ltd. is a commercial bank with multiple branches serving individual and business customers. Every day, the bank processes thousands of financial transactions through branch counters, ATMs, mobile banking, internet banking, and agency banking services. These transactions are stored in the bank's information system to support daily operations, customer service, and financial reporting.

Despite collecting large volumes of transaction data, the bank has not established a routine descriptive reporting process. Recent internal reviews have revealed missing records, duplicate transactions, inconsistent transaction descriptions, and formatting errors that reduce the reliability of management reports.

As a newly recruited Junior Data Analyst, you have been assigned to analyze the banking transaction data using Python. Before performing any analysis, you are required to assess data quality, clean the dataset, and produce descriptive summaries and visualizations that help management better understand customer transaction behavior.

Business Problem

Management requires answers to the following questions:

- Which branches process the highest number of transactions?
- Which banking channels are most frequently used?
- Which transaction types occur most often?
- What is the average transaction amount?
- Which customer age groups perform the highest transaction volumes?
- How do transaction patterns vary across months?
- Are there duplicate or inconsistent transaction records affecting reporting?
- How does transaction activity differ across regions?

Without accurate descriptive analysis, the bank cannot effectively monitor operational performance, allocate staff, improve customer service, or support strategic planning.

Importance of the Analysis

This analysis will help the bank to:

- Improve transaction data quality.
- Understand customer banking behavior.
- Compare branch performance.
- Evaluate banking channel utilization.
- Identify transaction trends.

- Produce accurate operational reports.
- Support evidence-based financial management.

Expected Outcome

Management expects a comprehensive analytical report containing:

- Branch performance summaries
- Transaction type analysis
- Banking channel usage statistics
- Customer demographic analysis
- Monthly transaction trends
- Descriptive statistical summaries
- Professional charts and dashboards -Recommendations for improving banking operations

Capstone Project 3: Agricultural Crop Production Analysis

ProjectLink: https://drive.google.com/file/d/1BjKDaDchCeTEmr30w_0f_0WBuk0fW0hr/view?usp=drive_link

DatasetLink: https://drive.google.com/file/d/1IuCnlKHuz1bb9b6ViAaWl9M39euyOpP_/view?usp=drive_link

Project Scenario

Background

Green Harvest Farmers' Cooperative is an agricultural organization that supports farmers across several farming regions. The cooperative collects production data on crop yields, farm sizes, fertilizer application, rainfall, irrigation methods, and market prices to monitor agricultural productivity and guide resource allocation.

Although thousands of farm production records have been collected over multiple growing seasons, the cooperative has not performed a comprehensive analysis to understand production trends or compare farming performance across regions. Preliminary inspections indicate that the dataset contains missing values, duplicate records, inconsistent crop names, and incomplete production information.

You have been appointed as a Junior Data Analyst to conduct a descriptive analysis of the agricultural production data using Python. Your task is to clean the dataset, explore production patterns, create informative visualizations, and prepare an analytical report that supports agricultural planning.

Business Problem

The cooperative management seeks answers to the following questions:

- Which crops produce the highest yields?
- Which farming regions record the highest production?
- What is the average farm size?
- How much fertilizer is applied to different crops?
- Which irrigation methods are most commonly used?
- How does rainfall vary across farming regions?
- What seasonal production patterns can be observed?
- Are there inconsistencies in the production records that need correction?

Without proper descriptive analysis, management cannot effectively distribute agricultural inputs, monitor productivity, or support farmers with evidence-based recommendations.

Importance of the Analysis

The analysis will enable the cooperative to:

- Improve agricultural data quality.
- Compare crop production across regions.
- Evaluate farming practices.
- Examine rainfall and fertilizer usage.
- Summarize seasonal production patterns.
- Produce statistical summaries and visualizations.
- Support informed agricultural planning and resource allocation.

Expected Outcome

The final report should provide:

- Regional crop production summaries
- Crop performance comparisons
- Farm size distribution
- Fertilizer and irrigation analyses
- Seasonal production trends
- Descriptive statistical summaries
- Professional visualizations
- Practical recommendations for improving agricultural productivity

WAS THIS HELPFUL?

https://cdam.chuka.ac.ke

Thank you

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